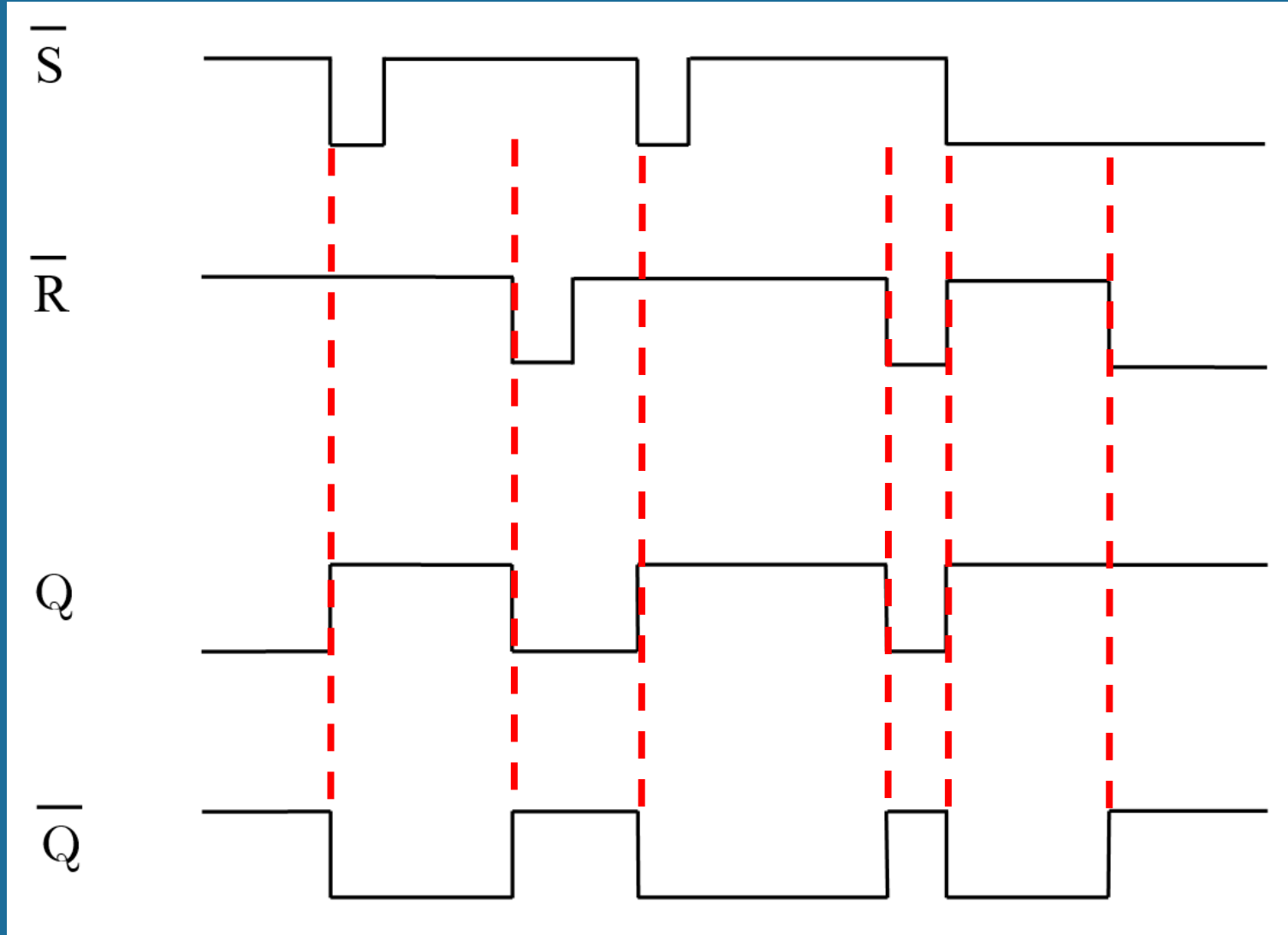
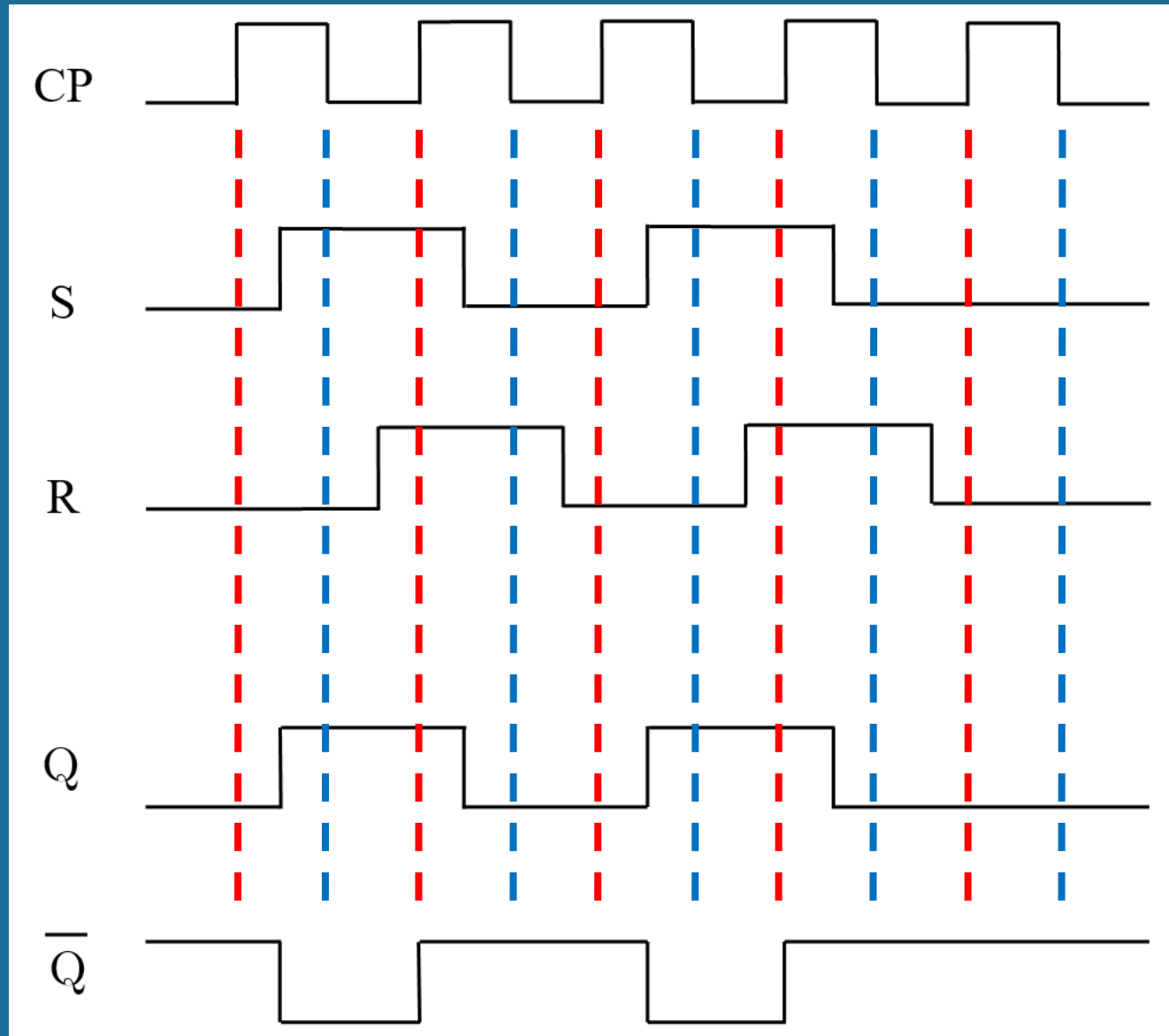


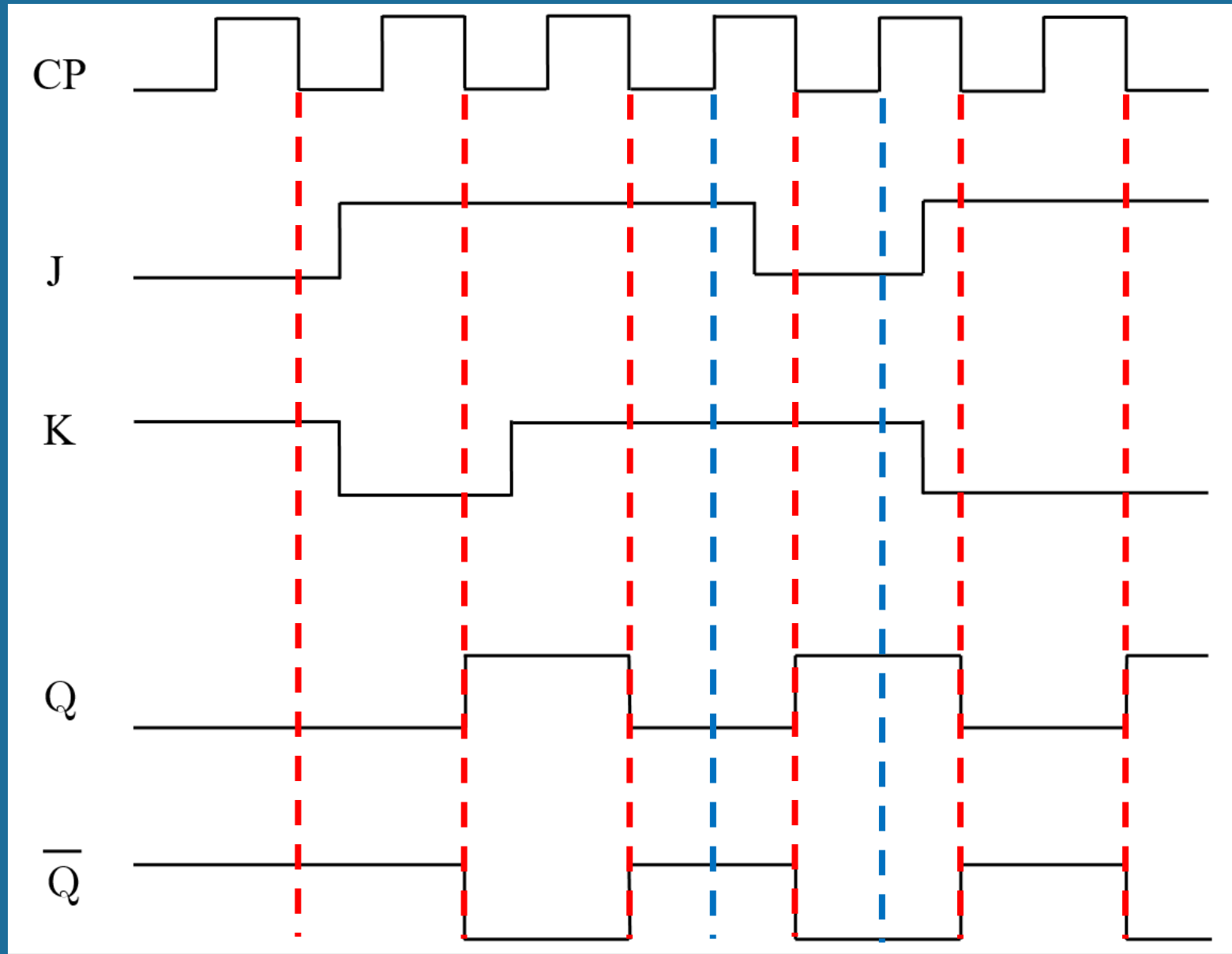
5.1 R-S latch composed of “NAND” gates



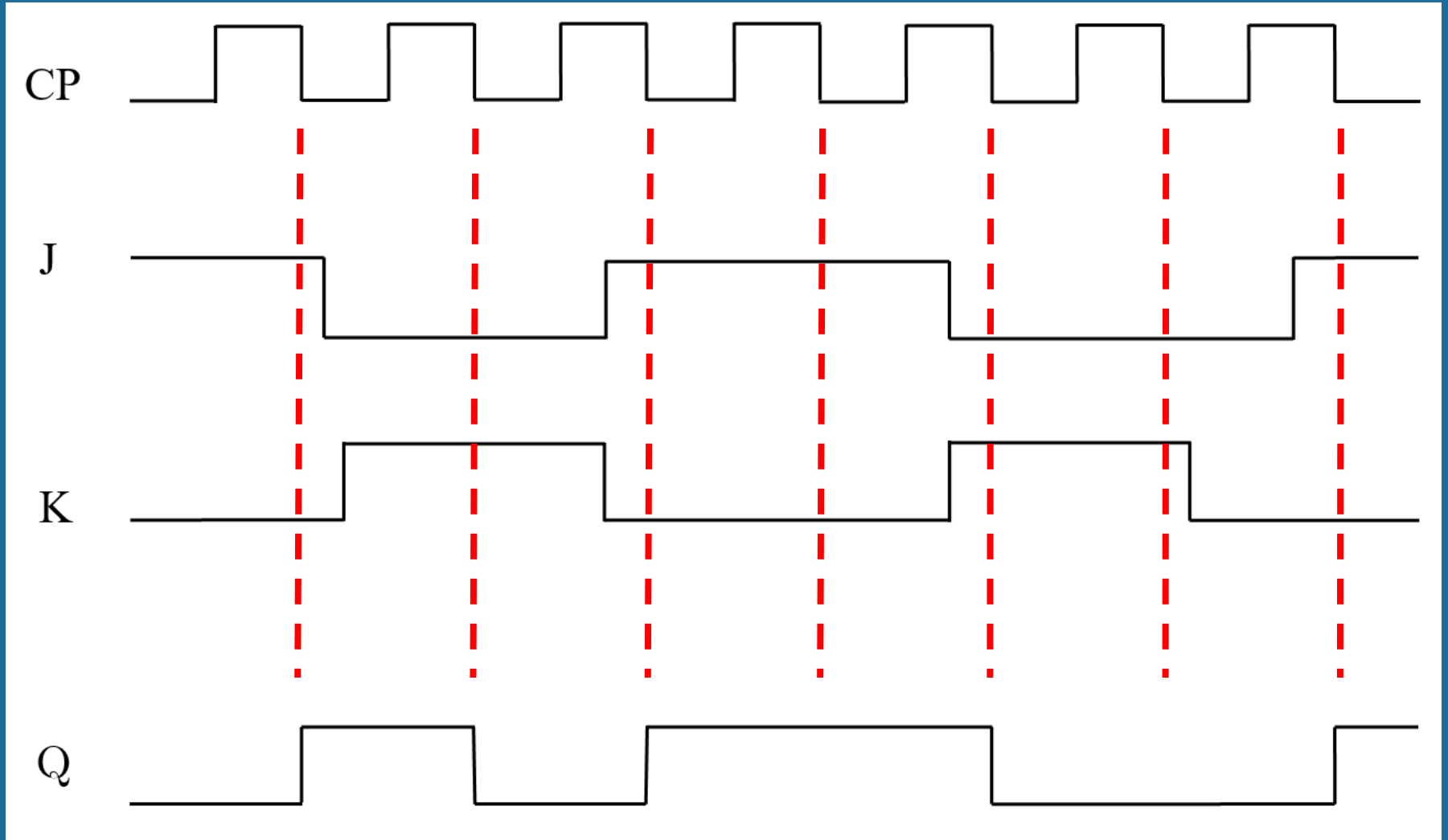
5.2 Gated R-S latch



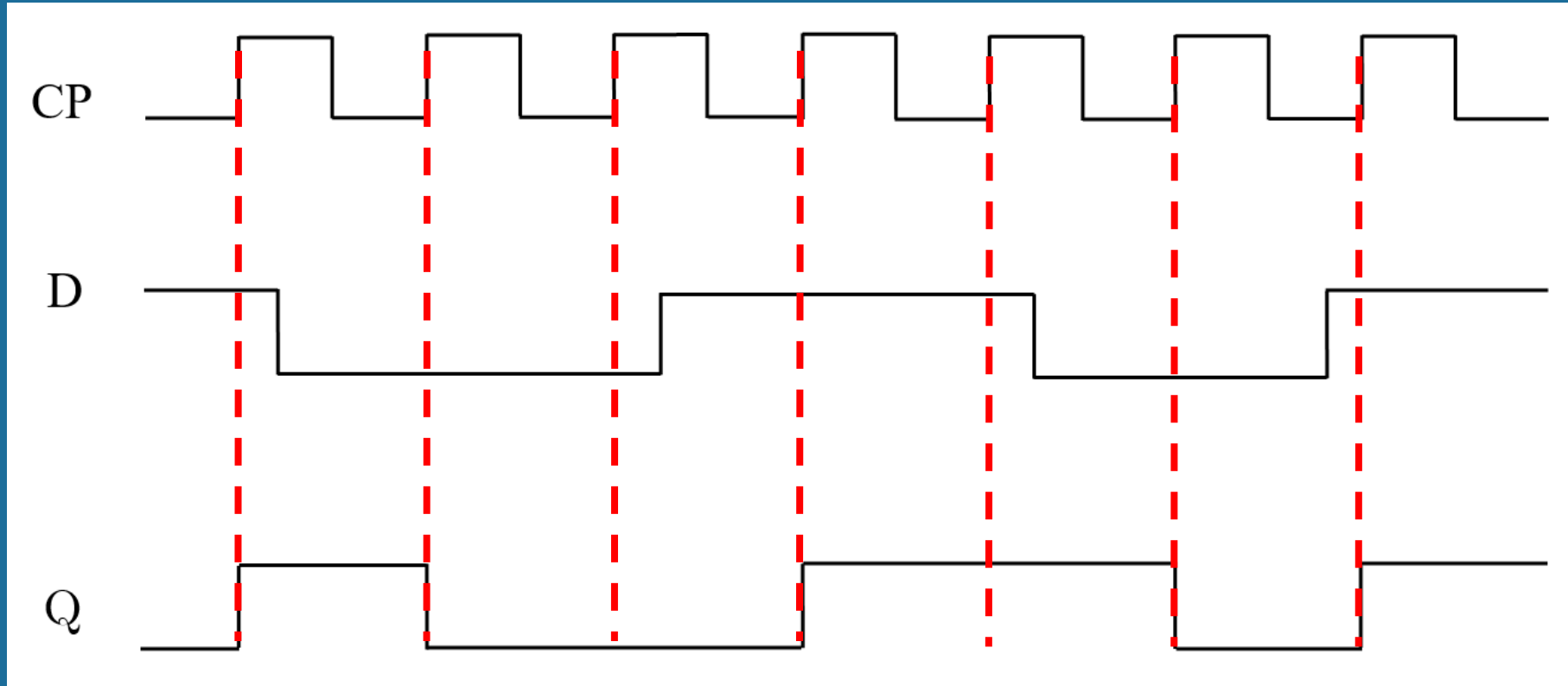
5.3 Master/slave J-K flip-flop



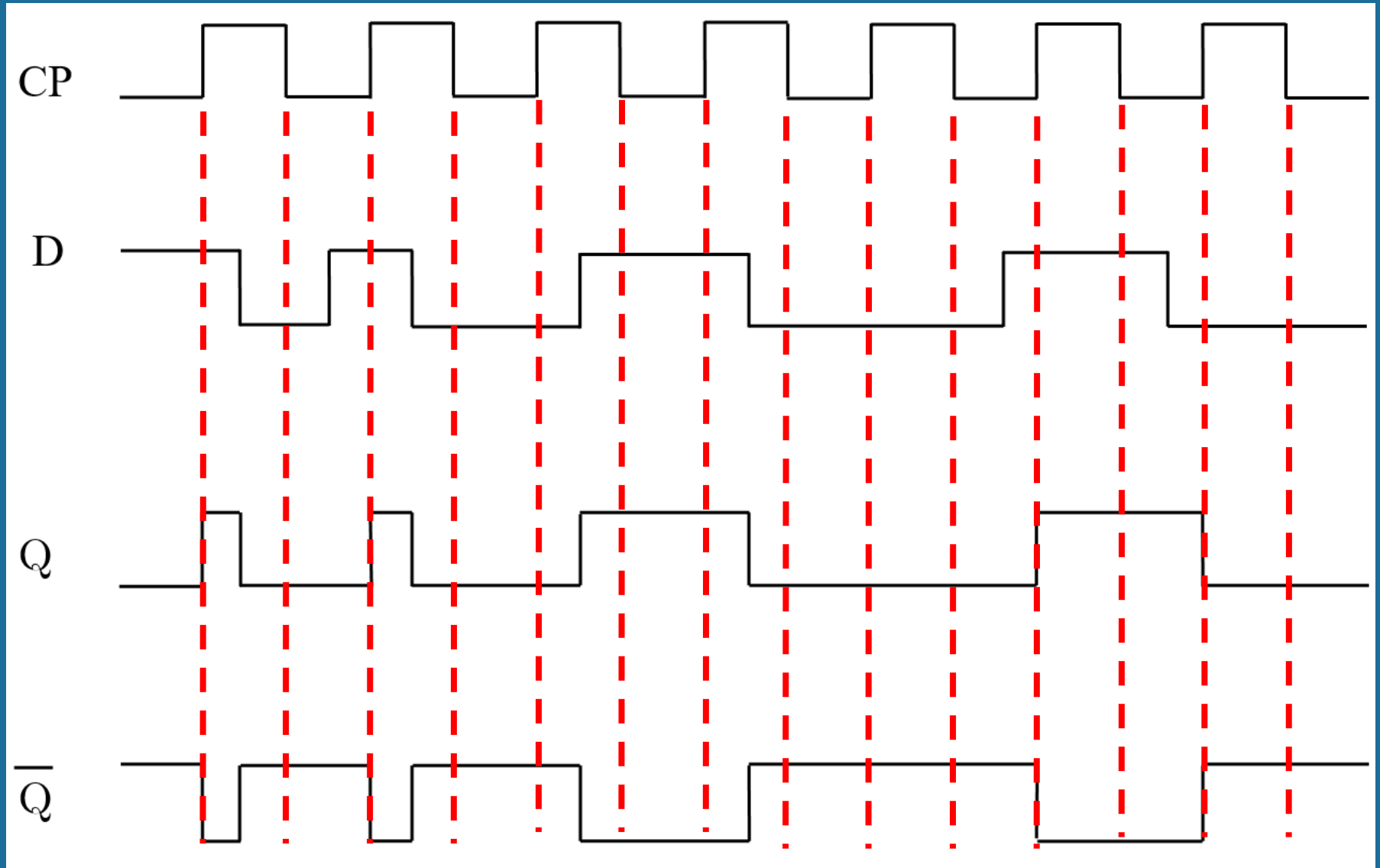
5.5 Falling-edge triggered J-K flip-flop



5.6 Rising-edge triggered D flip-flop

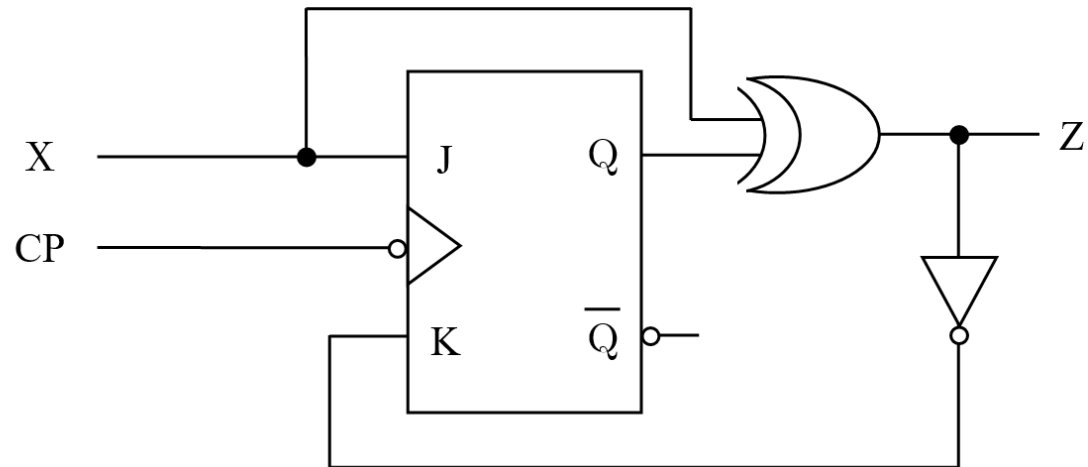


5.7 Gated D latch



6.1

Analyze the synchronous sequential logic circuit in Figure X6-1. (1) Write the output equation, the excitation equation, and the state equation. (2) Write the state transition truth table and draw the state transition diagram. (3) Describe the role of circuit.



Excitation Equation

$$J = X$$

$$K = \overline{X \oplus Q^n}$$

Output Equation

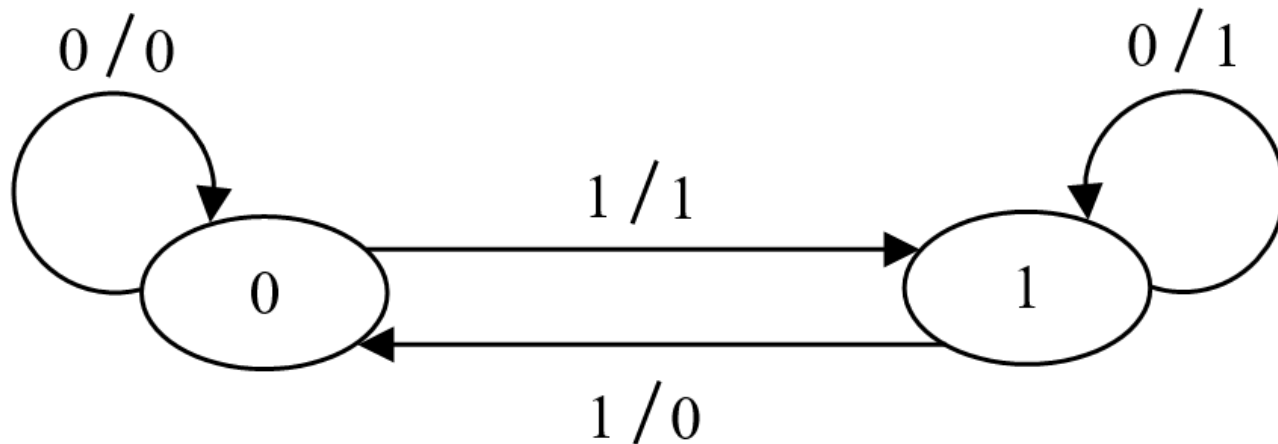
$$Z = X \oplus Q^n$$

State Equation

$$\begin{aligned} Q^{n+1} &= \overline{J}Q^n + \overline{K}Q^n \\ &= \overline{X}Q^n + (X \oplus Q^n)Q^n \\ &= \overline{X}Q^n + \overline{X}Q^n \\ &= X \oplus Q^n \end{aligned}$$

State Transition Truth Table of Exercise 6.1

X	Q^n	Q^{n+1}	Z
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

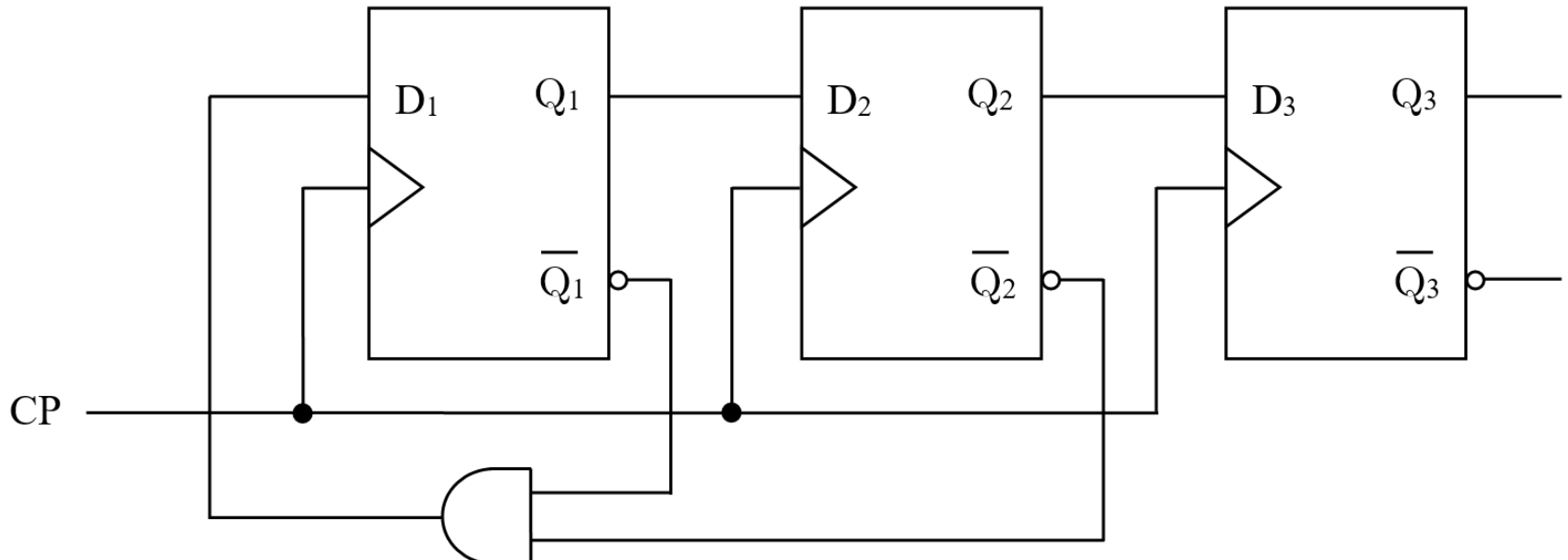


State Transition Diagram of Exercise 6.1

Role of Circuit: The circuit implements the “XOR” operation of the input variable “X” and the state variable “Q”.

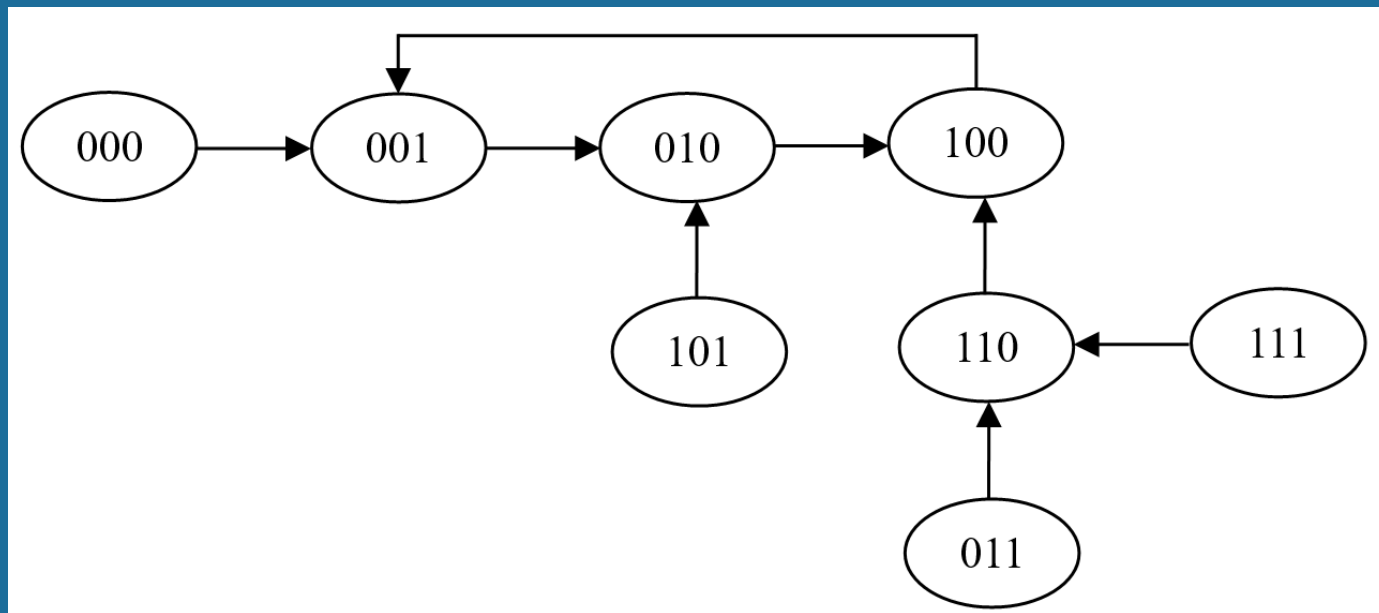
6.2

Analyze the synchronous sequential logic circuit in Figure X6-2. (1) Write the excitation equation and the state equation. (2) Write the state transition truth table and draw the state transition diagram. (3) Describe the role of circuit.



Excitation Equation	$D_1 = \overline{Q_1^n} \cdot \overline{Q_2^n}$	$D_2 = Q_1^n$	$D_3 = Q_2^n$
State Equation	$Q_1^{n+1} = \overline{Q_1^n} \cdot \overline{Q_2^n}$	$Q_2^{n+1} = Q_1^n$	$Q_3^{n+1} = Q_2^n$

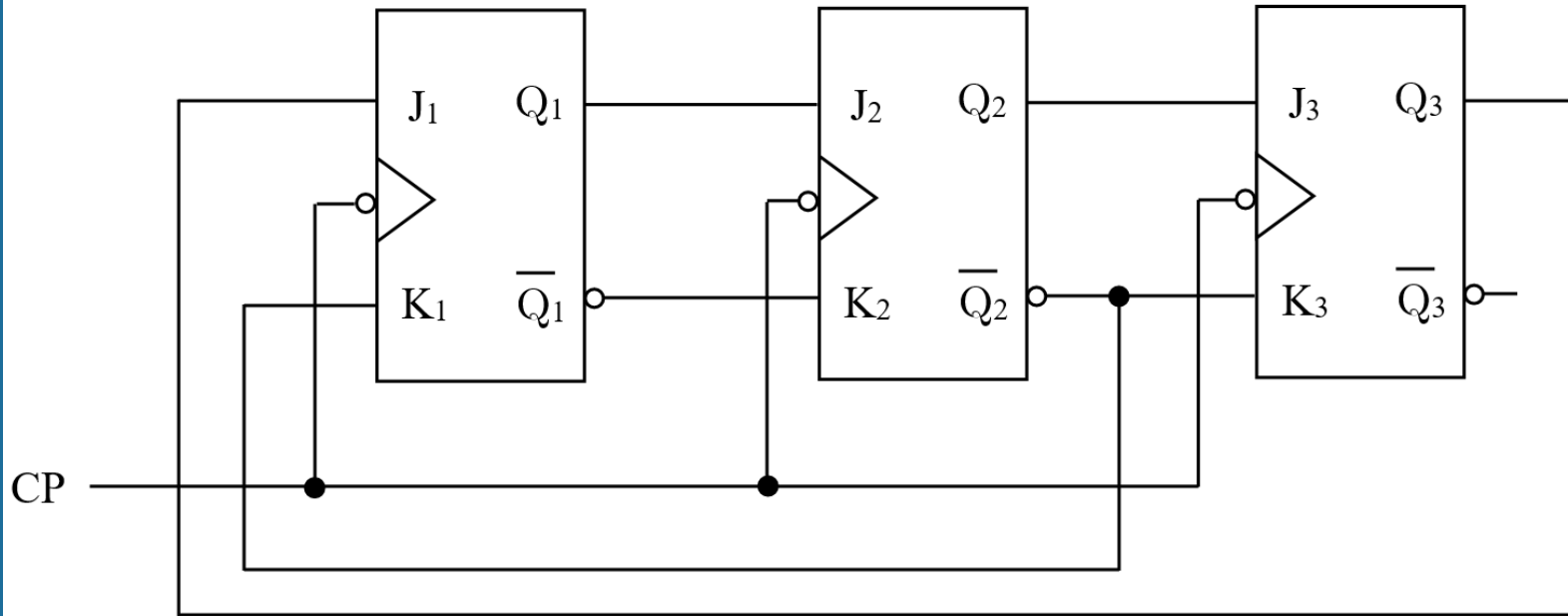
Q_3^n	Q_2^n	Q_1^n				Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}
0	0	0				0	0	1
0	0	1				0	1	0
0	1	0				1	0	0
0	1	1				1	1	0
1	0	0				0	0	1
1	0	1				0	1	0
1	1	0				1	0	0
1	1	1				1	1	0



Role of Circuit: The circuit is a unidirectional shift register, which is self-starting.

6.3

Analyze the synchronous sequential logic circuit in Figure X6-3. (1) Write the excitation equation and the state equation. (2) Write the state transition truth table and draw the state transition diagram. (3) Describe the role of circuit.



Excitation Equation

$$J_1 = Q_3^n$$

$$K_1 = \overline{Q_2^n}$$

$$J_2 = Q_1^n$$

$$K_2 = \overline{Q_1^n}$$

$$J_3 = Q_2^n$$

$$K_3 = \overline{Q_2^n}$$

Excitation Equation

$$J_1 = Q_3^n$$

$$K_1 = \overline{Q_2^n}$$

$$J_2 = Q_1^n$$

$$K_2 = \overline{Q_1^n}$$

$$J_3 = Q_2^n$$

$$K_3 = \overline{Q_2^n}$$

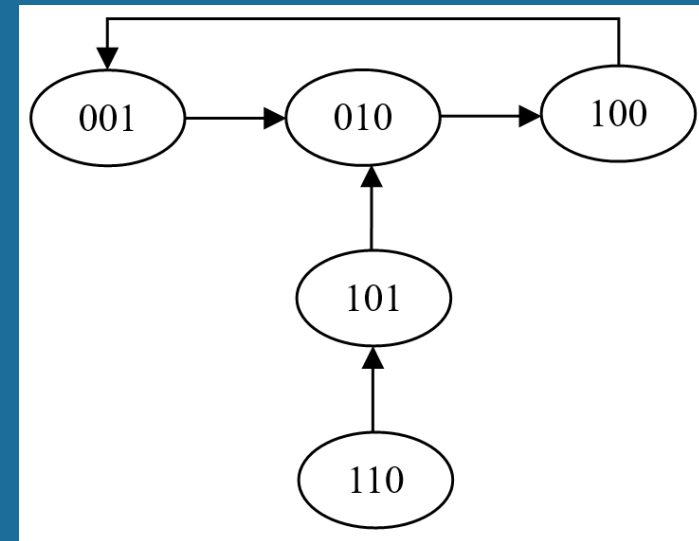
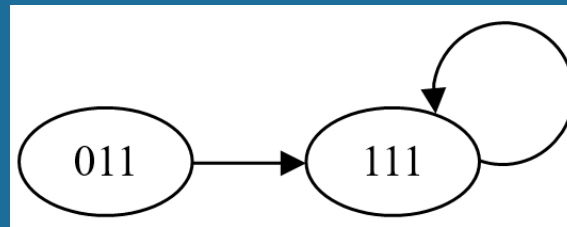
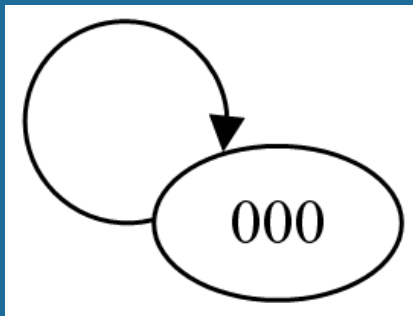
State Equation

$$Q_1^{n+1} = Q_3^n \overline{Q_1^n} + Q_2^n Q_1^n$$

$$Q_2^{n+1} = Q_1^n \overline{Q_2^n} + Q_1^n Q_2^n = Q_1^n$$

$$Q_3^{n+1} = Q_2^n \overline{Q_3^n} + Q_2^n Q_3^n = Q_2^n$$

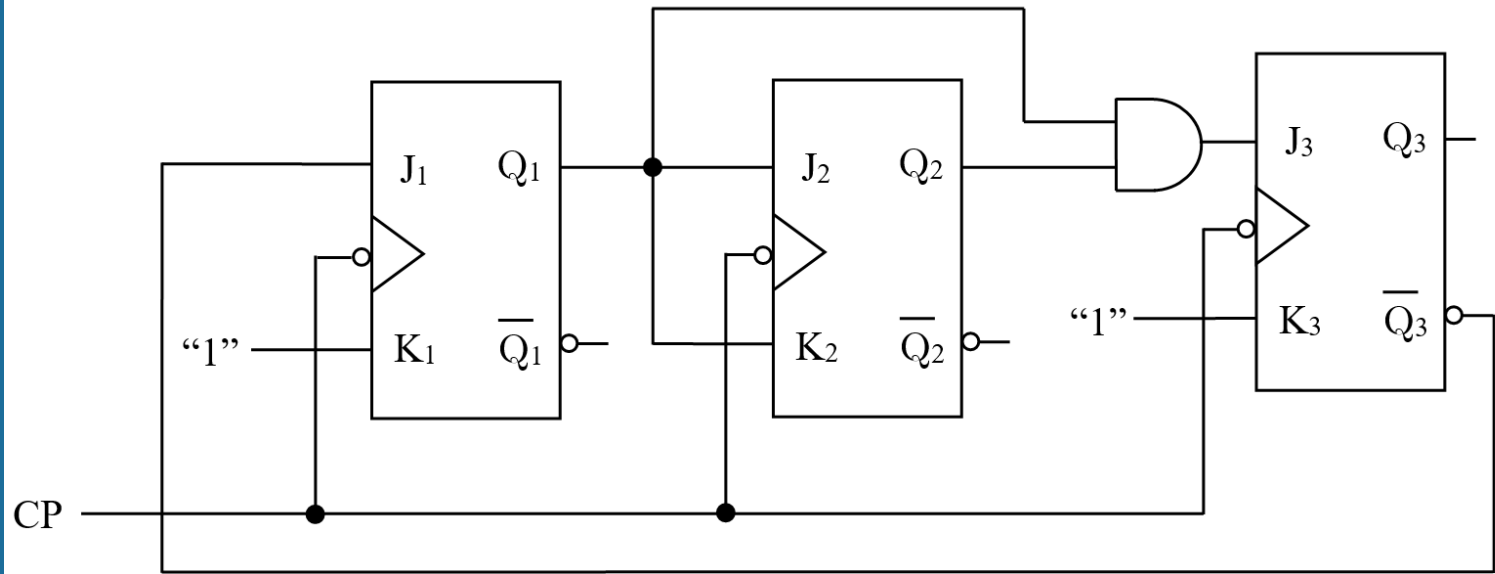
State Transition Truth Table of Exercise 6.3					
Q_3^n	Q_2^n	Q_1^n	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	1	1	1
1	0	0	0	0	1
1	0	1	0	1	0
1	1	0	1	0	1
1	1	1	1	1	1



Role of Circuit: The circuit is a unidirectional shift register, which cannot be self-starting.

6.4

Analyze the synchronous sequential logic circuit in Figure X6-4. (1) Write the excitation equation and the state equation. (2) Write the state transition truth table and draw the state transition diagram. (3) Describe the role of circuit.



$$J_1 = \overline{Q_3^n}$$

$$K_1 = 1$$

$$J_2 = Q_1^n$$

$$K_2 = Q_1^n$$

$$J_3 = Q_1^n \cdot Q_2^n$$

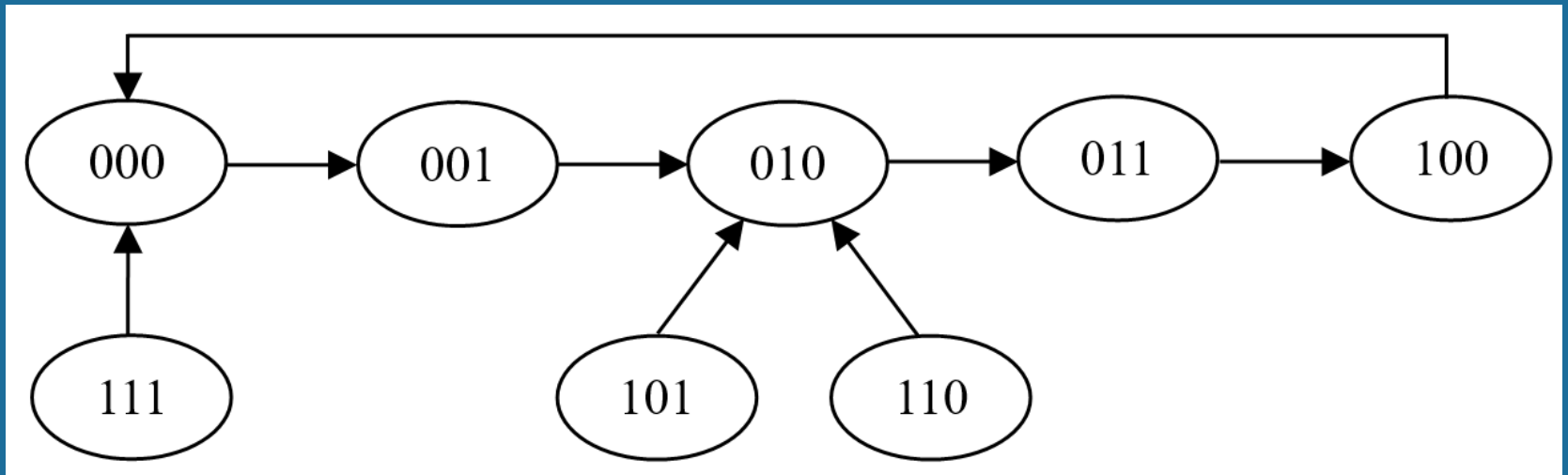
$$K_3 = 1$$

$$Q_1^{n+1} = \overline{Q_3^n} \cdot \overline{Q_1^n}$$

$$Q_2^{n+1} = Q_1^n \oplus Q_2^n$$

$$Q_3^{n+1} = Q_1^n Q_2^n \overline{Q_3^n}$$

Q_3^n	Q_2^n	Q_1^n	Q^{n+1}		
Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}			
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	0	1	0
1	1	0	0	1	0
1	1	1	0	0	0



Role of Circuit: The circuit is an addition counter. It counts five numbers.